

microsoft/mstic: RapidReleaseTI/MachineKeyScan.ps1

microsoft/mstic: RapidReleaseTI/MachineKeyScan.ps1 - microsoft, GitHub.

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Content

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MachineKeyScan.ps1

`microsoft/mstic` at `master`, path `RapidReleaseTI/MachineKeyScan.ps1`.

```

# This script is provided as is by Microsoft as intended to allow the detection of
publicly disclosed machine key element values
# within web.config files that are used by the IIS Web Server on Windows Server on
Windows Client operating systems.
# The script can also scan arbitrary .config files for the presence of the <machineKey>
element inside the <system.Web> attribute
# and detect if the values contained in the file are matching publicly disclosed ones.

# usage:
# run MachineKey-Scan.ps1 - to allow for scanning of all web.config files that are in
usage by the IIS web-server
# run MachineKey-Scan.ps1 -ConfigFile <pathToConfigFile> to scan an arbitrary
configuration file for the presence of
# publicly disclosed key values.

# version 1.2

param(
    [string]$ConfigFile # Optional: User-provided config file
)

# Fetch Microsoft-identified publicly disclosed MachineKeys.txt
$DisclosedKeysUrl =
"https://github.com/microsoft/mstic/blob/master/RapidReleaseTI/MachineKeys.csv"

try {
    $DisclosedKeys = Invoke-WebRequest -Uri $DisclosedKeysUrl -ErrorAction Stop -
UseBasicParsing
    if (-not $DisclosedKeys -or -not $DisclosedKeys.Content) {
        Write-Host -ForegroundColor Yellow "Error: Downloaded content from
$DisclosedKeysUrl is empty."
        exit 1
    }
    Write-Host -ForegroundColor Green "DisclosedKeys loaded successfully."
} catch {
    Write-Host -ForegroundColor Red "Error downloading MachineKeys.txt: $_"
    exit 1
}

# Load the WebAdministration module
Import-Module WebAdministration

# Method to get the .NET Framework installation folder

```

```

function Get-DotNetFrameworkFolder {
    param (
        [string]$version,
        [string]$architecture
    )

    Write-Host -ForegroundColor Yellow "`n*** Checking for .Net Framework installation
version: ($version) and architecture ($architecture)"

    if ($architecture -eq "64-bit") {
        $registryPath = "HKLM:\SOFTWARE\Microsoft\NET Framework Setup\NDP\$version\Full"
    } else {
        $registryPath = "HKLM:\SOFTWARE\WOW6432Node\Microsoft\NET Framework
Setup\NDP\$version\Full"
    }

    $installPath = (Get-ItemProperty -Path $registryPath -Name InstallPath -ErrorAction
SilentlyContinue).InstallPath

    #return $installPath
    if ($installPath) {
        return $installPath
    } else {
        Write-Host -ForegroundColor Yellow "`tThe specified .NET Framework version
($version) is not installed."
        return $null
    }
}

```

Define the method allowing to check the machineKey element for known disclosed key values

```

function Check-MachineKey {
    param (
        [string]$webConfigFilePathEnv,
        [string]$hashedKeyContents,
        [int]$tabCount
    )

    $tabs = "`t" * $tabCount
    write-host "${tabs}Checking $webConfigFilePathEnv"
    [xml]$webConfig = Get-Content $webConfigFilePathEnv

```

```

    # Check if the machineKey element exists and check if either the validationKey or
    decryptionKey is public
    if ($webConfig.configuration.'system.web'.machineKey)
    {
        $machineKey = $webConfig.configuration.'system.web'.machineKey
        $validationKey = $machineKey.validationKey
        $decryptionKey = $machineKey.decryptionKey

        if ($validationKey) {
            $hashValidationKey =
[BitConverter]::ToString([System.Security.Cryptography.SHA256]::Create().ComputeHash([Sys
tem.Text.Encoding]::UTF8.GetBytes($validationKey.ToUpper())) -replace "-", ""
            if ($hashedKeyContents -match $hashValidationKey) {
                Write-Host -ForegroundColor Red "${tabs}!!!WARNING!!!: validationKey
$validationKey - is publicly disclosed!"
            }
        }

        if ($decryptionKey) {
            $hashDecryptionKey =
[BitConverter]::ToString([System.Security.Cryptography.SHA256]::Create().ComputeHash([Sys
tem.Text.Encoding]::UTF8.GetBytes($decryptionKey.ToUpper())) -replace "-", ""
            if ($hashedKeyContents -match $hashDecryptionKey) {
                Write-Host -ForegroundColor Red "${tabs}!!!WARNING!!!: decryptionKey
$decryptionKey - is publicly disclosed!"
            }
        }
    }
}

# Define the method to check for web applications inside a site
function Check-WebApplications {
    param (
        [string]$siteName,
        [string]$hashedKeyContents
    )

    $webApps = Get-WebApplication -Site $siteName
    $webAppCount = $webApps.Count
    write-host -ForegroundColor Cyan "`tFound $webAppCount web application(s)"
    if ($webAppCount -gt 0) {
        foreach ($webApp in $webApps) {
            write-host "`tName: $($webApp.Path), Root Folder: $($webApp.PhysicalPath)"
        }
    }
}

```

```

        $webAppPhysicalPathEnv =
[Environment]::ExpandEnvironmentVariables($webApp.PhysicalPath)
        $webConfigFiles = Get-ChildItem -Path "$($webAppPhysicalPathEnv)" -Recurse -
Filter "web.config"
        foreach ($webConfigFile in $webConfigFiles) {
            $webConfigFilePath = $webConfigFile.FullName
            $webConfigFilePathEnv =
[Environment]::ExpandEnvironmentVariables($webConfigFilePath)
            Check-MachineKey -webConfigFilePathEnv $webConfigFilePathEnv -
hashedKeyContents $hashedKeyContents -tabCount 2
        }
    }
}

```

Define the method to check for virtual directories inside a site

```

function Check-VirtualDirectories {
    param (
        [string]$siteName,
        [string]$hashedKeyContents
    )

    $virtualDirs = Get-WebVirtualDirectory -Site $siteName
    $virtualDirCount = $virtualDirs.Count
    write-host -ForegroundColor Cyan "`tFound $virtualDirCount virtual directory(ies)"
    if ($virtualDirCount -gt 0) {
        foreach ($virtualDir in $virtualDirs) {
            write-host "`tName: $($virtualDir.Path), Root Folder:
$($virtualDir.PhysicalPath)"
            $virtualDirPhysicalPathEnv =
[Environment]::ExpandEnvironmentVariables($virtualDir.PhysicalPath)
            $webConfigFiles = Get-ChildItem -Path "$($virtualDirPhysicalPathEnv)" -
Recurse -Filter "web.config"
            foreach ($webConfigFile in $webConfigFiles) {
                $webConfigFilePath = $webConfigFile.FullName
                $webConfigFilePathEnv =
[Environment]::ExpandEnvironmentVariables($webConfigFilePath)
                Check-MachineKey -webConfigFilePathEnv $webConfigFilePathEnv -
hashedKeyContents $hashedKeyContents -tabCount 2
            }
        }
    }
}

```

```

# Method to process a web.config file
function Check-WebConfigFile {
    param([string]$WebConfigFilePath,
        [string]$hashedKeyContents)

    try {
        [xml]$WebConfig = Get-Content -Path $WebConfigFilePath -ErrorAction Stop
    } catch {
        Write-Host -ForegroundColor Yellow "Error: Unable to read $WebConfigFilePath.
Skipping."
        return
    }

    # invoke the code to check the file directly
    Check-MachineKey -webConfigFilePathEnv $WebConfigFilePath -hashedKeyContents
$hashedKeyContents -tabCount 1
}

# If user provides a config file, process only that file
if ($ConfigFile) {
    if (-not (Test-Path $ConfigFile)) {
        Write-Host -ForegroundColor Red "Error: Specified config file '$ConfigFile' does
not exist."
        exit 1
    }
    Write-Host "Checking user-provided config file: $ConfigFile"
    Check-WebConfigFile -WebConfigFilePath $ConfigFile -hashedKeyContents
$DisclosedKeys.Content
    exit 0
}

# Get the Installation of .Net Framework (version 2 and 4, 32 and 64 bits)
$dotNet2Folder32 = Get-DotNetFrameworkFolder -version "v2.0.50727" -architecture "32-bit"
$dotNet2Folder64 = Get-DotNetFrameworkFolder -version "v2.0.50727" -architecture "64-bit"

$dotNet4Folder32 = Get-DotNetFrameworkFolder -version "v4" -architecture "32-bit"
$dotNet4Folder64 = Get-DotNetFrameworkFolder -version "v4" -architecture "64-bit"

# Check if we have the Framework installed, check the config files
if ($dotNet2Folder32) {
    Check-MachineKey -webConfigFilePathEnv ($dotNet2Folder32 + "\Config\machine.config")
-hashedKeyContents $DisclosedKeys.Content -tabCount 1
}

```

```

}

if ($dotNet2Folder64) {
    Check-MachineKey -webConfigFilePathEnv ($dotNet2Folder64 + "\Config\machine.config")
    -hashedKeyContents $DisclosedKeys.Content -tabCount 1
}

if ($dotNet4Folder32) {
    Check-MachineKey -webConfigFilePathEnv ($dotNet4Folder32 + "\Config\machine.config")
    -hashedKeyContents $DisclosedKeys.Content -tabCount 1
}

if ($dotNet4Folder64) {
    Check-MachineKey -webConfigFilePathEnv ($dotNet4Folder64 + "\Config\machine.config")
    -hashedKeyContents $DisclosedKeys.Content -tabCount 1
}

# Get list of IIS sites
$sites=Get-ChildItem IIS:\Sites
# Iterate through each website
foreach ($site in $sites)
{
    write-host -ForegroundColor DarkYellow "`n***** Site: $($site.Name)"
    $physicalPath = [Environment]::ExpandEnvironmentVariables($site.PhysicalPath)
    $webConfigFiles = Get-ChildItem -Path "$physicalPath" -Recurse -Filter "web.config"
    # Iterate web.config files for each site

    foreach ($webConfigFile in $webConfigFiles) {
        # Load the XML content of the web.config file
        $webConfigFilePath=$webConfigFile.FullName
        $webConfigFilePathEnv=
[Environment]::ExpandEnvironmentVariables($webConfigFilePath)

        Check-MachineKey -webConfigFilePathEnv $webConfigFilePathEnv -hashedKeyContents
$DisclosedKeys.Content -tabCount 1

    }

    # Check for all web applications inside a site
    Check-WebApplications -siteName $site.Name -hashedKeyContents $DisclosedKeys.Content

    # Check for all virtual directories inside a site
    Check-VirtualDirectories -siteName $site.Name -hashedKeyContents

```

```
$DisclosedKeys.Content
```

```
}
```

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